

DSE5002 - Project #1

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Project Objectives: The goal of this project is to analyze dataset comprised of international and domestic salaries for all levels of data scientists, who can lead a team and be paid competitively.

Questions to Address:

1. What does the data set look like?
2. What is the salary difference being offshore vs domestic employees?
3. What does the salary range look like for full-time data scientist who may eventually lead the team?
4. What is competitive salary for this role?
5. Does the company size affect salary range?
6. Does experience affect the salary range?
7. How much experience should our candidate have?
8. What does salary trend look like for these roles?

LLM Support: Utilized ChatGPT to help refine the questions

In [1]: *#Libraries:*

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import os
```

In [296... salary_df=pd.read_csv("project_1_data.csv")

In [297... salary_df.head(5)

Out [297...

	Unnamed: 0	work_year	experience_level	employment_type	job_title	salary	salary_in_usd
0	0	2020	MI	FT	Data Scientist	70000	70000.000000
1	1	2020	SE	FT	Machine Learning Scientist	260000	260000.000000
2	2	2020	SE	FT	Big Data Engineer	85000	85000.000000
3	3	2020	MI	FT	Product Data Analyst	20000	20000.000000
4	4	2020	SE	FT	Machine Learning Engineer	150000	150000.000000

Question #1: What does the data set look like?

In [298...

```
salary_df.shape
```

Out [298...

(607, 12)

In [299...

```
salary_df.describe()
```

Out [299...

	Unnamed: 0	work_year	salary	salary_in_usd	remote_ratio
count	607.000000	607.000000	6.070000e+02	607.000000	607.000000
mean	303.000000	2021.405272	3.240001e+05	112297.869852	70.92257
std	175.370085	0.692133	1.544357e+06	70957.259411	40.70913
min	0.000000	2020.000000	4.000000e+03	2859.000000	0.000000
25%	151.500000	2021.000000	7.000000e+04	62726.000000	50.00000
50%	303.000000	2022.000000	1.150000e+05	101570.000000	100.00000
75%	454.500000	2022.000000	1.650000e+05	150000.000000	100.00000
max	606.000000	2022.000000	3.040000e+07	600000.000000	100.00000

In [300...

```
salary_df.dtypes
```

```
Out[300...] Unnamed: 0          int64
work_year          int64
experience_level    object
employment_type     object
job_title           object
salary             int64
salary_currency     object
salary_in_usd       int64
employee_residence  object
remote_ratio        int64
company_location    object
company_size        object
dtype: object
```

```
In [301...] salary_df.isnull().sum()
```

```
Out[301...] Unnamed: 0          0
work_year          0
experience_level    0
employment_type     0
job_title           0
salary             0
salary_currency     0
salary_in_usd       0
employee_residence  0
remote_ratio        0
company_location    0
company_size        0
dtype: int64
```

```
In [302...] # Analysis of duplicates:
```

```
salary_df.duplicated().sum()
```

```
Out[302...] np.int64(0)
```

```
In [303...] #Verification of Unnamed being a variable or not:
```

```
salary_df["Unnamed: 0"].head(5)
```

```
Out[303...] 0    0
1    1
2    2
3    3
4    4
Name: Unnamed: 0, dtype: int64
```

```
In [304...] # Removing Unnamed:0
```

```
salary_df = salary_df.drop(columns=["Unnamed: 0"])
```

```
salary_df.head()
```

Out [304...

	work_year	experience_level	employment_type	job_title	salary	salary_currency
0	2020	MI	FT	Data Scientist	70000	EUR
1	2020	SE	FT	Machine Learning Scientist	260000	USD
2	2020	SE	FT	Big Data Engineer	85000	GBP
3	2020	MI	FT	Product Data Analyst	20000	USD
4	2020	SE	FT	Machine Learning Engineer	150000	USD

In [305...

```
# Plot - Histogram of Salaries in USD to show the distribution:
```

```
plt.hist(salary_df["salary_in_usd"])

plt.title("Salary Distribution Across the Full Dataset")
plt.xlabel("Salary in USD")
plt.ylabel("Frequency")
plt.show()
```



Observations:

1. The dataset has 607 rows and 12 columns
2. The data covers years 2020 through 2022
3. Mean salary = ~\$112,297 USD, Min salary = \$2,859 USD, Max salary = \$600,000 USD, Median salary = \$101,570 USD.
4. Maximum salary is much higher than 75% percentile (\$600k vs \$150k)
5. The data has both numeric and categorical variables
6. No missing data in the dataset
7. Unnamed is an index column not a variable.

Question #2: What is the salary difference being offshore vs domestic employees?

In [306... `salary_df.head()`

Out [306...

	work_year	experience_level	employment_type	job_title	salary	salary_currency
0	2020	MI	FT	Data Scientist	70000	EUR
1	2020	SE	FT	Machine Learning Scientist	260000	USD
2	2020	SE	FT	Big Data Engineer	85000	GBP
3	2020	MI	FT	Product Data Analyst	20000	USD
4	2020	SE	FT	Machine Learning Engineer	150000	USD

In [307... `salary_df["employee_residence"].value_counts()`

Out[307... employee_residence

US	332
GB	44
IN	30
CA	29
DE	25
FR	18
ES	15
GR	13
JP	7
PT	6
PK	6
BR	6
NL	5
IT	4
RU	4
PL	4
AE	3
TR	3
AU	3
VN	3
AT	3
DK	2
NG	2
HU	2
MX	2
SI	2
RO	2
BE	2
SG	2
PH	1
CN	1
HN	1
NZ	1
UA	1
IQ	1
CL	1
MT	1
IR	1
CO	1
HR	1
BG	1
KE	1
MD	1
RS	1
HK	1
LU	1
JE	1
CZ	1
PR	1
AR	1
DZ	1
MY	1
TN	1
EE	1
BO	1

```
IE      1  
CH      1  
Name: count, dtype: int64
```

```
In [308... #Mean salaries by country:  
  
salary_df.groupby(["employee_residence"])["salary_in_usd"].mean()
```

```
Out[308... employee_residence
AE      100000.000000
AR       60000.000000
AT       76738.666667
AU      108042.666667
BE       85699.000000
BG       80000.000000
BO       75000.000000
BR       54634.666667
CA       97085.310345
CH      122346.000000
CL       40038.000000
CN       43331.000000
CO       21844.000000
CZ       69999.000000
DE       85552.560000
DK       37252.500000
DZ      100000.000000
EE       32974.000000
ES       57593.400000
FR       59886.611111
GB       81403.159091
GR       56331.230769
HK       66022.000000
HN       20000.000000
HR       45618.000000
HU       35997.000000
IE       71444.000000
IN       37322.333333
IQ      100000.000000
IR        4000.000000
IT       61600.000000
JE      100000.000000
JP      103537.714286
KE        9272.000000
LU       59102.000000
MD       18000.000000
MT       28369.000000
MX       18185.000000
MY      200000.000000
NG       30000.000000
NL       60956.600000
NZ      125000.000000
PH       45760.000000
PK       27462.833333
PL       56177.500000
PR      160000.000000
PT       42862.500000
RO       51419.000000
RS       25532.000000
RU      105750.000000
SG      104176.500000
SI       63831.000000
TN       31875.000000
TR       20096.666667
UA       13400.000000
```

```
US    149194.117470
VN     30800.000000
Name: salary_in_usd, dtype: float64
```

In [309... *# US salary ave (USD):*

```
us_salary = salary_df[salary_df["employee_residence"] == "US"]

us_salary["salary_in_usd"].mean()
```

Out[309... np.float64(149194.1174698795)

In [310... *# Offshore salary ave (USD)*

```
offshore_salary = salary_df[salary_df["employee_residence"] != "US"]

offshore_salary["salary_in_usd"].mean()
```

Out[310... np.float64(67754.03636363636)

In [311... *# Salary difference US - Offshore:*

```
salary_difference = us_salary["salary_in_usd"].mean() - offshore_salary["salary_in_usd"].mean()

salary_difference
```

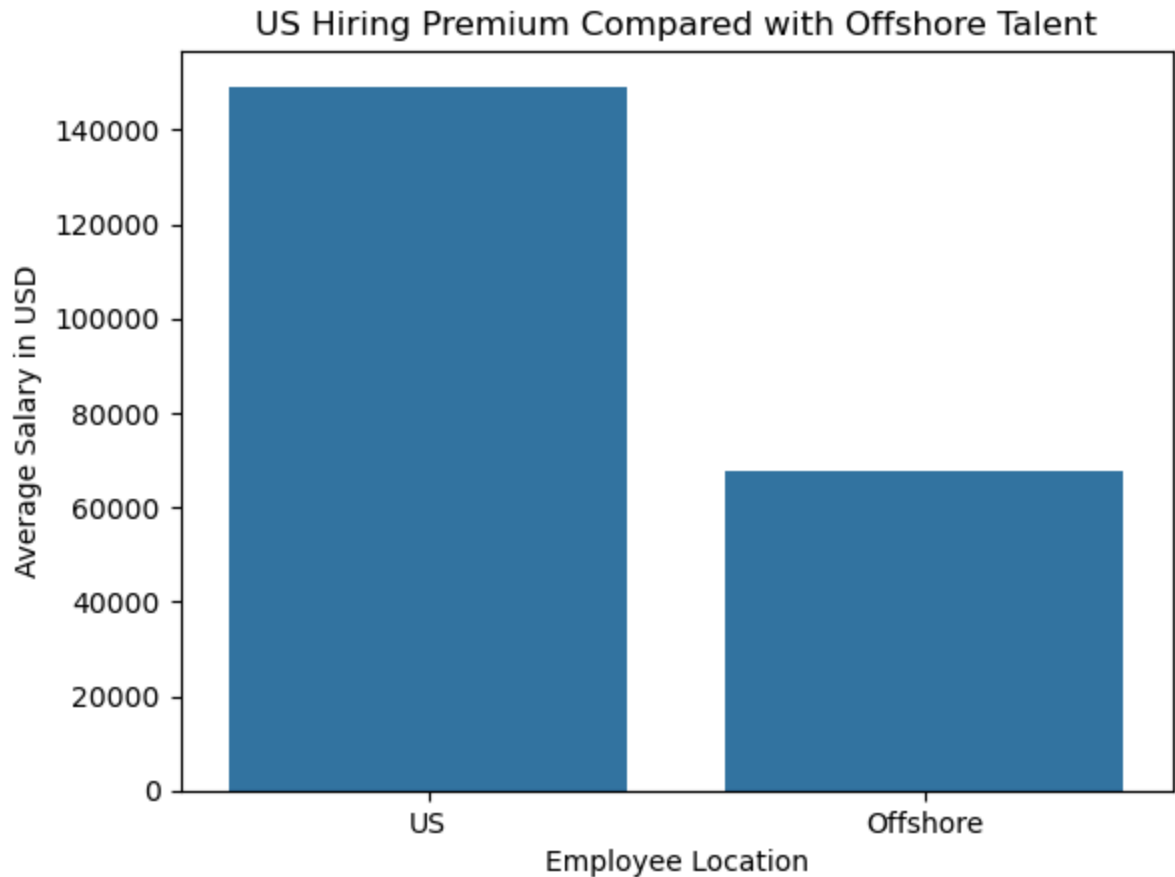
Out[311... np.float64(81440.08110624315)

In [312... *# Plot - Comparison of US vs Offshore salaries:*

```
location_salary = pd.DataFrame({
    "Location": ["US", "Offshore"],
    "Average_Salary": [
        us_salary["salary_in_usd"].mean(),
        offshore_salary["salary_in_usd"].mean()]})

sns.barplot(x="Location", y="Average_Salary", data=location_salary)

plt.title("US Hiring Premium Compared with Offshore Talent")
plt.xlabel("Employee Location")
plt.ylabel("Average Salary in USD")
plt.show()
```



Observations:

1. 332 US employees vs 275 Offshore employees
2. US salary ave (USD) = \$149,194, PR ave = \$160k, MY ave = \$200,000 (highest salary Note: PR and MY only have 1 data point, which is not comparable to US having 332 data points.
3. Offshore salary ave (USD) = ~ \$67,754
4. Salary delta (US vs Offshore in USD) = ~\$81,440 (us salaries are higher by \$81k)

Question #3: What does the salary range look like for full-time data scientist who may eventually lead the team?

In [313... *# Filtering by fulltime employee:*

```
ft_salary = salary_df[salary_df["employment_type"] == "FT"]
```

In [314... *# Filtering by data scientists:*

```
ft_data_scientist = ft_salary[ft_salary["job_title"] == "Data Scientist"]
```

In [315... *# Summary of salary distribution:*

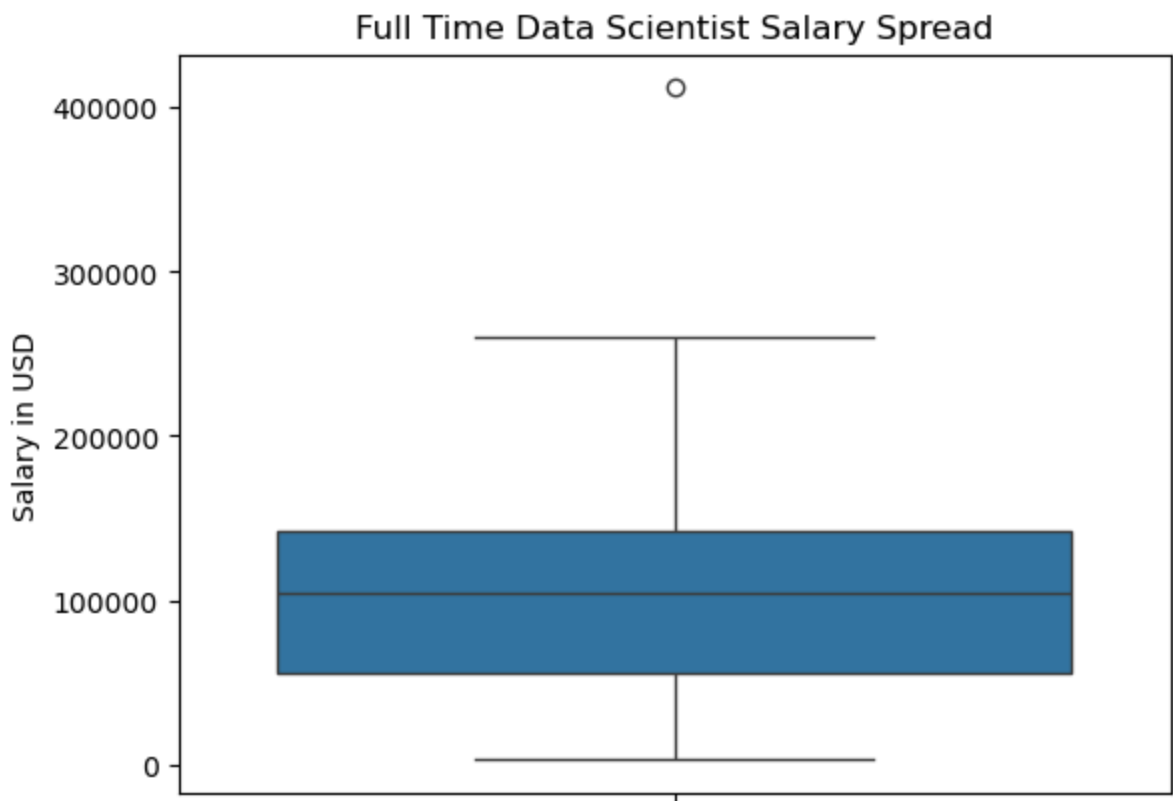
```
ft_data_scientist["salary_in_usd"].describe()
```

```
Out[315... count      140.000000
mean      108922.792857
std       64372.431676
min       2859.000000
25%      55490.000000
50%     104796.000000
75%     141975.000000
max      412000.000000
Name: salary_in_usd, dtype: float64
```

```
In [316... # Plot of full time data science salaries distribution:

sns.boxplot(y="salary_in_usd", data=ft_data_scientist)

plt.title("Full Time Data Scientist Salary Spread")
plt.ylabel("Salary in USD")
plt.show()
```



```
In [317... # Experience:

ft_data_scientist["experience_level"].value_counts()
```

```
Out[317... experience_level
SE      61
MI      59
EN      20
Name: count, dtype: int64
```

```
In [318... # Salaries per experience level:
```

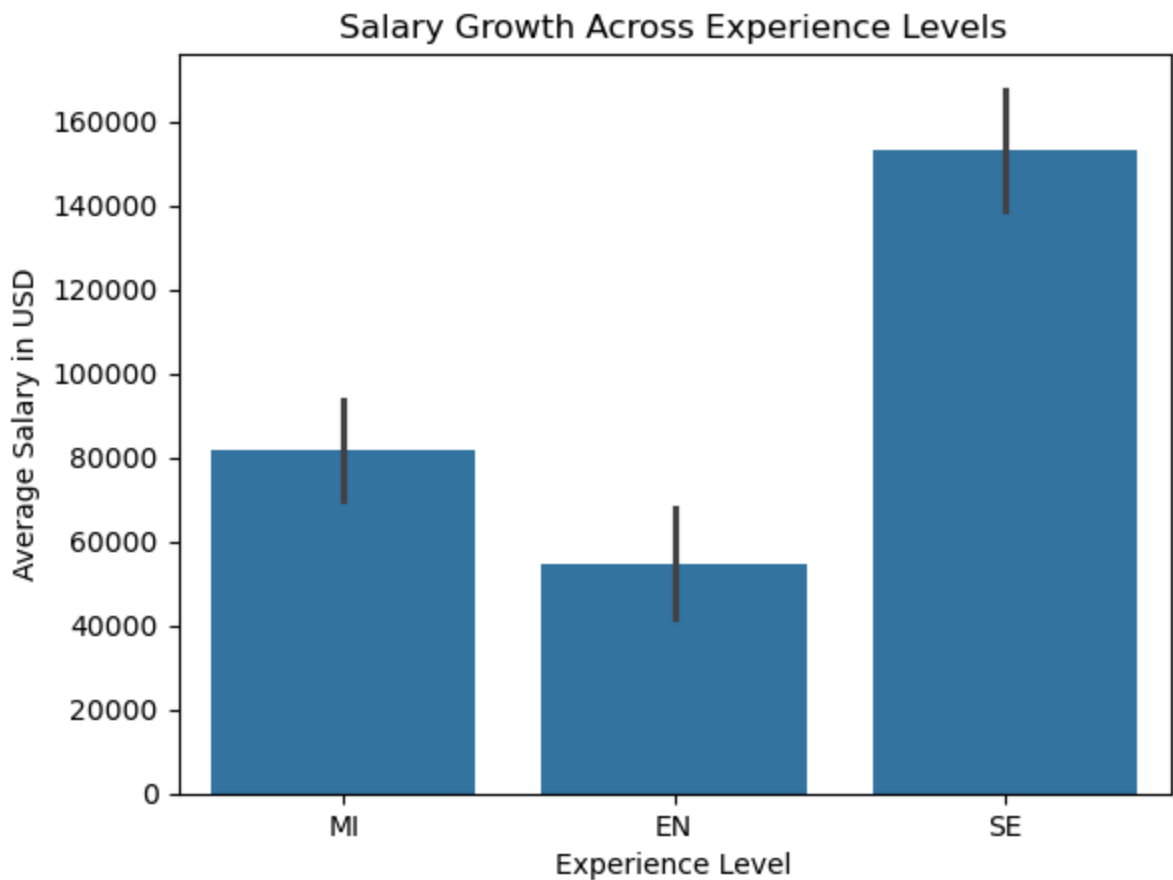
```
ft_data_scientist.groupby(["experience_level"])["salary_in_usd"].mean()
```

```
Out[318]: experience_level
EN      54780.550000
MI      81734.711864
SE     152971.016393
Name: salary_in_usd, dtype: float64
```

```
In [319]: # Plot of Ave salaries vs experience:

sns.barplot(
    x="experience_level",
    y="salary_in_usd",
    data=ft_data_scientist)

plt.title("Salary Growth Across Experience Levels")
plt.xlabel("Experience Level")
plt.ylabel("Average Salary in USD")
plt.show()
```



Observations (USD):

1. There are 140 data scientists data points.
2. Min salary = \$2,859, Max salary = 412,000, Mean salary = ~108,922, 75% = 141,975
3. There are 61 senior, 59 middle level and 20 entry level data points for fulltime data scientists.
4. Entry level roles earn = ~ \$54,780, Middle = ~ \$81,734, Senior = ~\$152,971

Question #4: What is competitive salary for this role?

```
In [320... # Analyzing senior roles only ("eventually lead the team"):
```

```
senior_data_scientist = ft_data_scientist[  
    ft_data_scientist["experience_level"] == "SE"]
```

```
In [321... # Filter by US only:
```

```
us_senior_data_scientist = senior_data_scientist[  
    senior_data_scientist["employee_residence"] == "US"]
```

```
In [322... # Data summary:
```

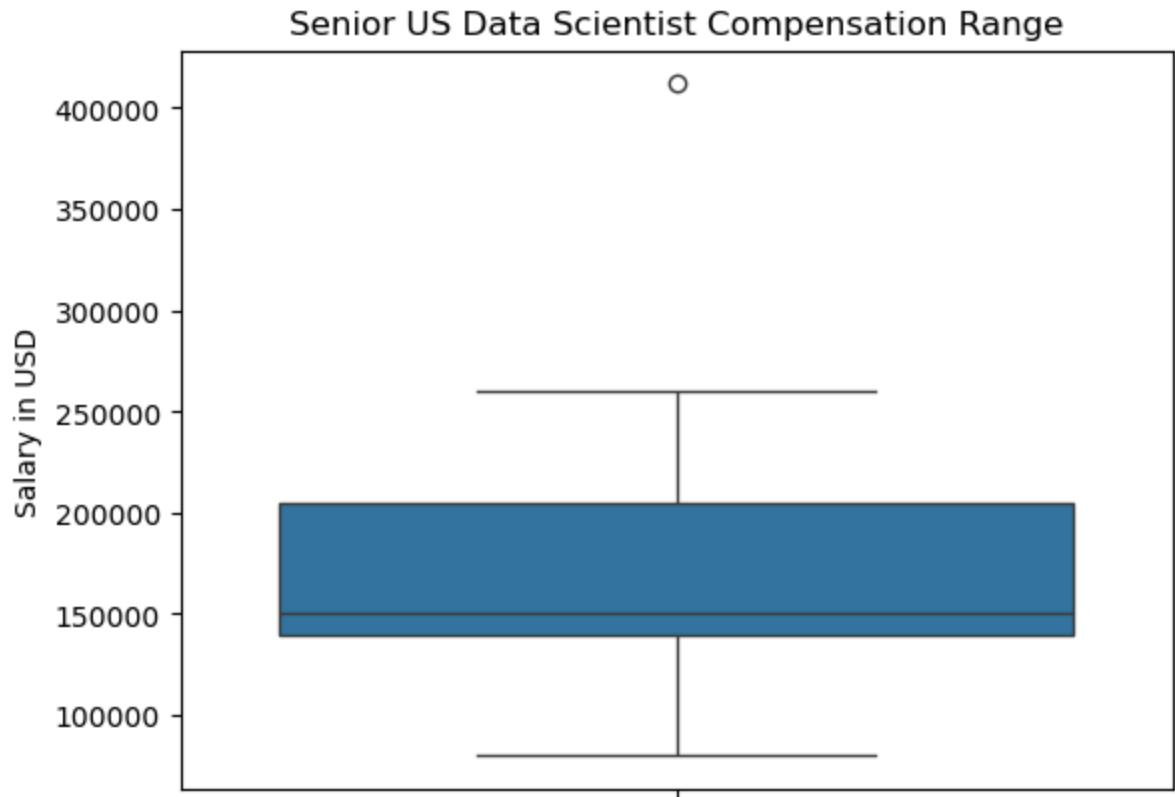
```
us_senior_data_scientist["salary_in_usd"].describe()
```

```
Out[322... count          51.000000  
mean       166796.078431  
std        53241.525488  
min         80000.000000  
25%       139300.000000  
50%       150000.000000  
75%       205300.000000  
max        412000.000000  
Name: salary_in_usd, dtype: float64
```

```
In [323... # Plot of Sr level salary distribution in US:
```

```
sns.boxplot(y="salary_in_usd", data=us_senior_data_scientist)
```

```
plt.title("Senior US Data Scientist Compensation Range")  
plt.ylabel("Salary in USD")  
plt.show()
```



Observations (USD):

1. There are 51 full time senior level roles in US
2. Ave salary is ~ 166,796 for these roles
3. Min salary is \$80,000
4. Max salary is \$412,000
5. 75% is \$205,300 (potential recommended salary to be competitive)
6. Salary range is \$80,000 to \$412,000

Question #5: Does the company size affect salary range?

```
In [324...] us_senior_data_scientist["company_size"].value_counts()
```

```
Out[324...] company_size
M      39
L      12
Name: count, dtype: int64
```

```
In [325...] # Comparison of ave saleries for Medium to Large companies in US for Sr. dat
us_senior_data_scientist.groupby(["company_size"])["salary_in_usd"].mean()
```

```
Out[325...] company_size
L      183908.333333
M      161530.769231
Name: salary_in_usd, dtype: float64
```

```
In [326... # Expanding the analysis to all levels of data scientists to understand salary  
ft_data_scientist["company_size"].value_counts()
```

```
Out[326... company_size  
M      75  
L      45  
S      20  
Name: count, dtype: int64
```

```
In [327... # Ave salary across all company sizes for all data scientists:  
ft_data_scientist.groupby(["company_size"])["salary_in_usd"].mean()
```

```
Out[327... company_size  
L      103313.355556  
M      127084.240000  
S       53438.600000  
Name: salary_in_usd, dtype: float64
```

```
In [328... # Plot of ave salaries by company size:  
  
sns.barplot(  
    x="company_size",  
    y="salary_in_usd",  
    data=ft_data_scientist)  
  
plt.title("Full Time Data Scientist Pay by Company Size")  
plt.xlabel("Company Size")  
plt.ylabel("Average Salary in USD")  
plt.show()
```



Observations:

1. The dataset contains 39 people from medium size companies and 12 from large.
2. There is no data points for small size companies in the US (Sr. level roles), which makes it harder to predict what small companies would pay for the same talent.
3. Ave salary for Sr. data scientists in large companies is ~ \$183,908
4. Ave salary for Sr. data scientists in Medium companies is ~ \$161,530
5. After expending the analysis accross all company sizes for all data scientists roles, there are 20 data scientists in small companies, 75 in medium and 45 in large.
6. After expending the analysis accross all company sizes for all data scientists roles, ave salaries for small companies are ~\$53,438, medium ~\$127,084 and \$103,313 for large companies.
7. Large and Medium size compapanies pay mich more than at small organizations, with medium companies paying the most.

Question #6: Does experience affect the salary range?

```
In [329... # Comparison of salaries for all data scientists across different seniority
ft_data_scientist.groupby(["experience_level"])["salary_in_usd"].describe()
```

Out [329...

		count	mean	std	min	25%	50%	
experience_level								
	EN	20.0	54780.550000	30224.880120	4000.0	34920.0	50483.5	
	MI	59.0	81734.711864	47212.036738	2859.0	41339.0	76958.0	1
	SE	61.0	152971.016393	58985.402166	20171.0	120000.0	140400.0	16

Observations:

1. Entry data scientists on average earn ~\$54,780, with STD= ~\$30,224, Min= \$4,000, Max= \$105,000, and 75% = \$81,675
2. Mid data scientists on average earn ~\$81,734, with STD= ~\$47,212, Min= ~\$2,859, Max= \$200,000 and 75%= ~\$118,529.
3. Senior data scientists on average earn ~\$152,971, with STD= ~\$58,985, Min= ~20,171, Max= \$412,000 and 75%= \$180,000.

Question #7: How much experience should our candidate have?

In [330...

```
# Analysis of full time data scientists in US to help us fill the role:

us_ft_data_scientist = ft_data_scientist[
    ft_data_scientist["employee_residence"] == "US"]

us_ft_data_scientist["experience_level"].value_counts()
```

Out [330...

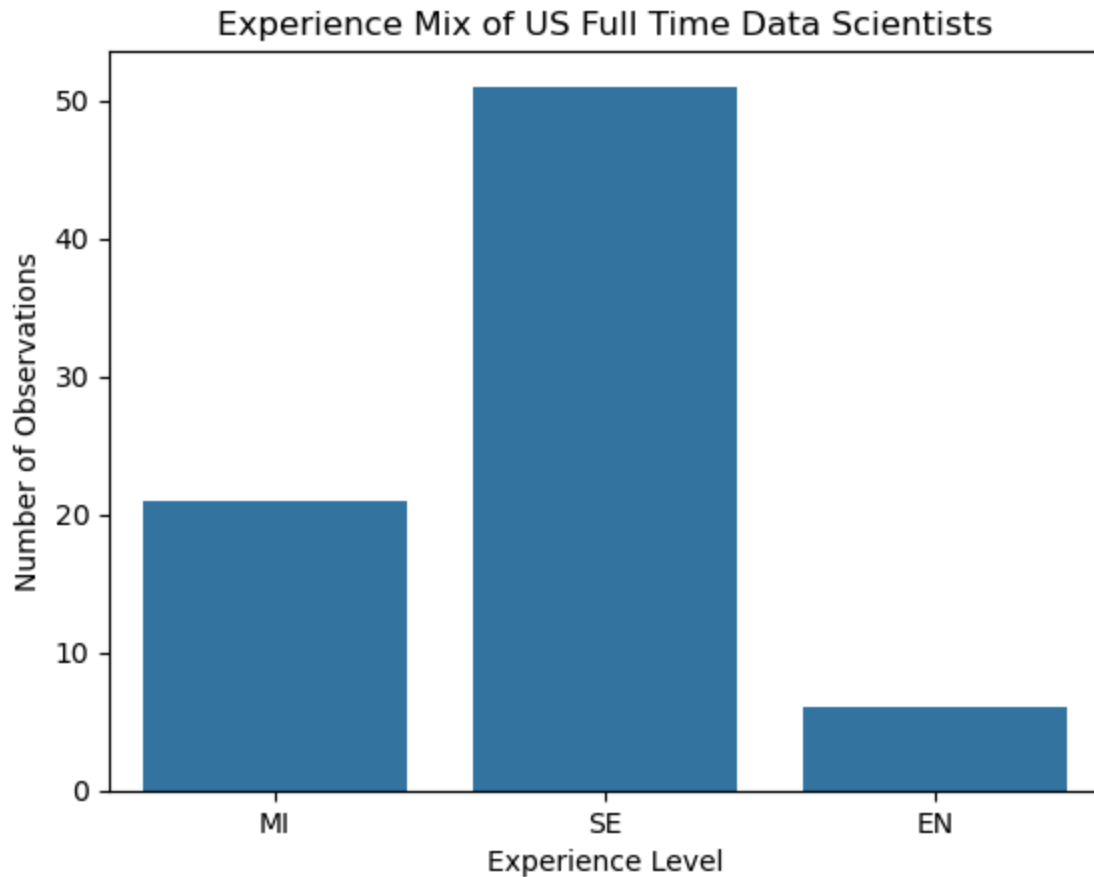
```
experience_level
SE      51
MI      21
EN       6
Name: count, dtype: int64
```

In [331...

```
# Plot of US data scientists by experience (full time)

sns.countplot(
    x="experience_level",
    data=us_ft_data_scientist
)

plt.title("Experience Mix of US Full Time Data Scientists")
plt.xlabel("Experience Level")
plt.ylabel("Number of Observations")
plt.show()
```



Observations:

1. We have 6 entry level roles, 21 medium and 51 senior level.
2. Sr. roles make up the majority of US market accross full time data scientists roles in this dataset.
3. Sr. roles is our target list of candidates that could lead the team and help us fill the role

Question #8: What does salary trend look like for these roles?

In [332... *# Analysis of data points per Sr. level in US per year:*

```
us_senior_data_scientist["work_year"].value_counts()
```

Out[332... work_year
 2022 47
 2020 2
 2021 2
 Name: count, dtype: int64

In [333... *# Analysis of data points for all levels of data scientists per year:*

```
ft_data_scientist["work_year"].value_counts()
```

```
Out[333...] work_year
2022      75
2021      45
2020      20
Name: count, dtype: int64
```

```
In [334...] # Analysis of all full time data scientists per year by salary (USD):

ft_data_scientist.groupby(["work_year"])["salary_in_usd"].mean()
```

```
Out[334...] work_year
2020      89185.600000
2021      70671.733333
2022     137136.680000
Name: salary_in_usd, dtype: float64
```

```
In [335...] # Salary yearly averages:

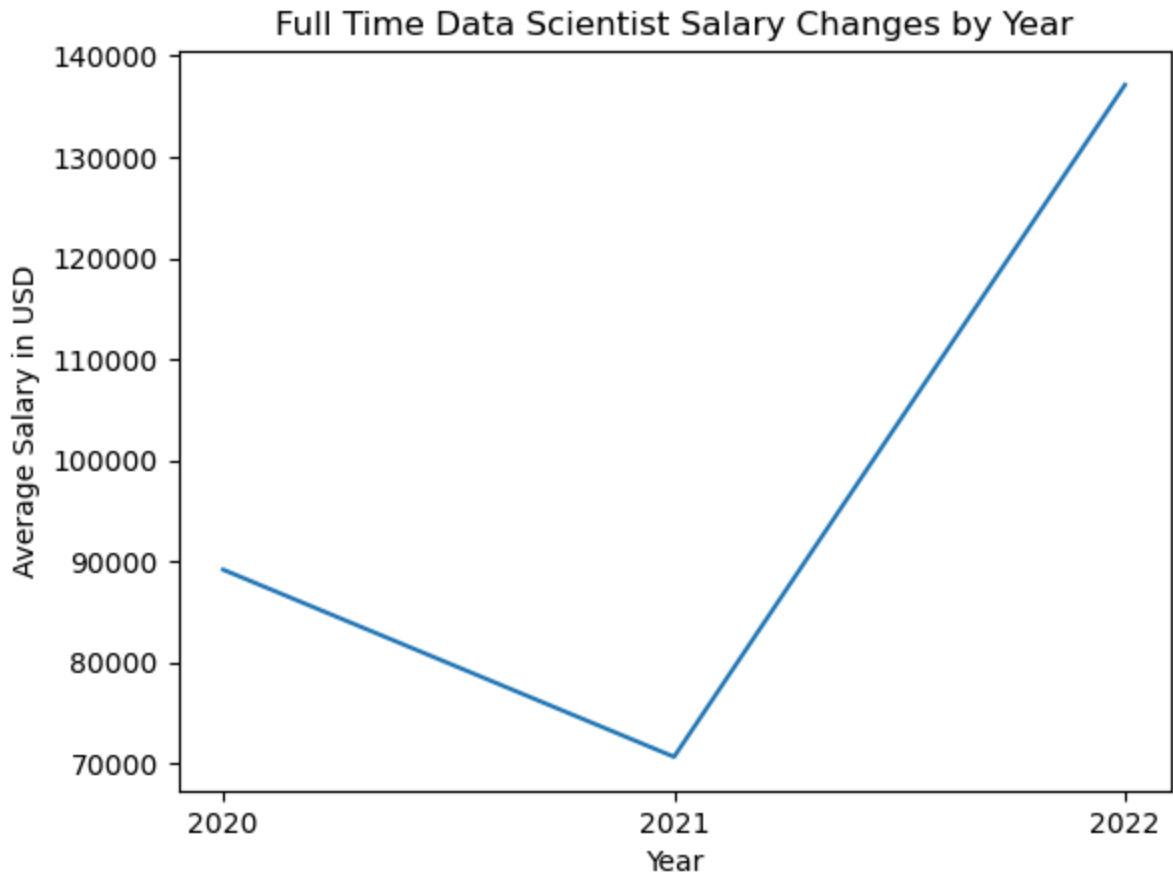
salary_by_year = ft_data_scientist.groupby(["work_year"])["salary_in_usd"].n

plt.plot(salary_by_year)

plt.title("Full Time Data Scientist Salary Changes by Year")
plt.xlabel("Year")
plt.ylabel("Average Salary in USD")

plt.xticks([2020, 2021, 2022])

plt.show()
```



Observaion:

1. For Sr. level roles, there's not enough data to create a meaningful trend, where years 2020 and 2021 only have 2 data points each.
2. For all levels data scientists, theres enough data to create a trend (2022=75 data points, 2021=45 data points and 2020=20 data points).
3. Salaries for full time data scientists is varied by year and doesnt show a linear trend (2020= ~\$89,185, 2021= ~\$70,671 and 2022 = ~\$137,136).

In [336... *# Analysis of experience level in small companies:*

```
small_company_ds = ft_data_scientist[
    ft_data_scientist["company_size"] == "S"]

small_company_ds["experience_level"].value_counts()
```

Out[336... experience_level

```
MI    12
EN     6
SE     2
Name: count, dtype: int64
```

In [337... *# Analysis of 2022 salaries for Sr. level roles of full time data scientists*

```
us_senior_2022 = us_senior_data_scientist[
    us_senior_data_scientist["work_year"] == 2022]
```

```
us_senior_2022["salary_in_usd"].describe()
```

```
Out [337... count      47.000000
mean    163289.361702
std      41132.338722
min       80000.000000
25%     140000.000000
50%     150000.000000
75%     205300.000000
max     260000.000000
Name: salary_in_usd, dtype: float64
```

Final Analysis Summary:

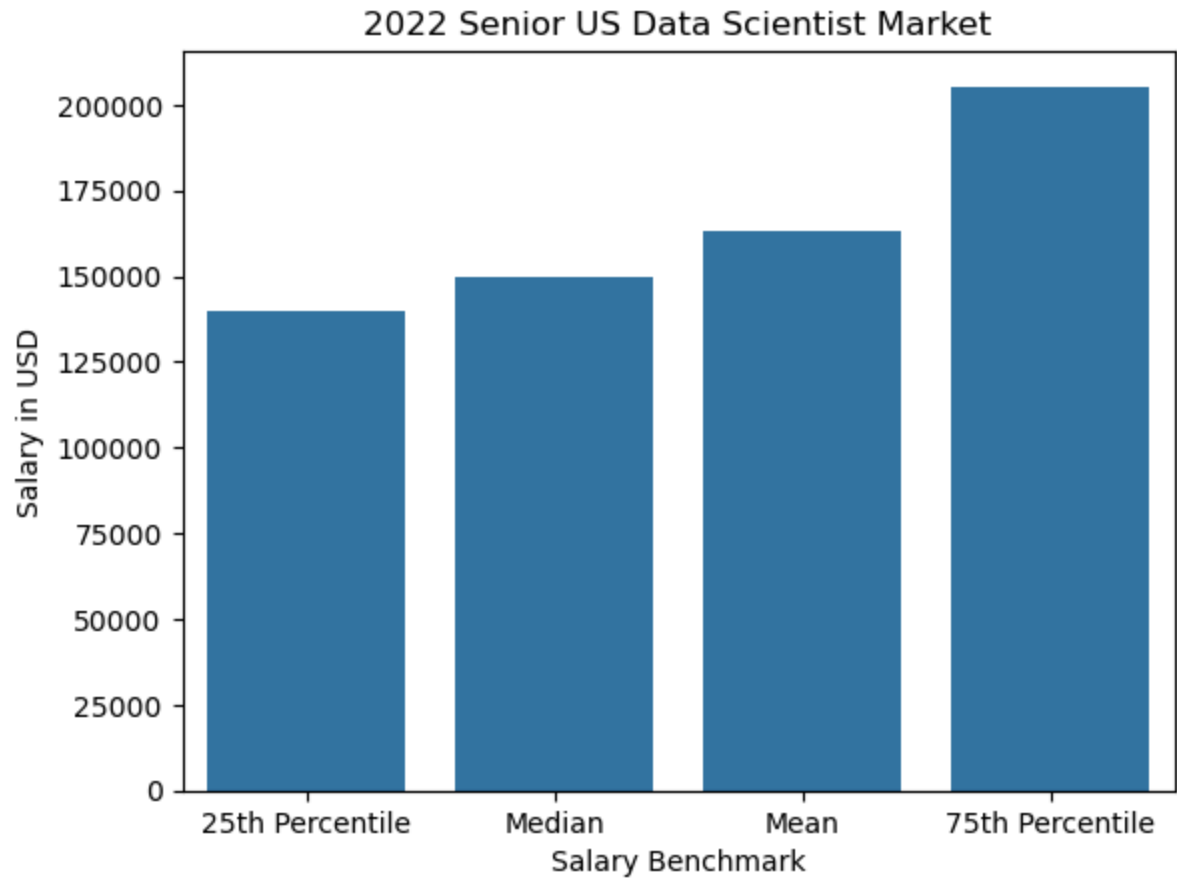
1. For US Sr. level fulltime Data Scientists, the median salary is \$150,000 and the 75% is ~\$205,300, which supports using these values from the market data.
2. 2022 data shows the same median and 75% (\$150,000 and \$205,300)
3. The salary average in small companies for full time data scientists are not comparable as there are only 2 data points for Sr level roles that we need.

```
In [338... # Plot of US Sr level data scientists in 2022:

salary_benchmarks = pd.DataFrame({
    "Benchmark": ["25th Percentile", "Median", "Mean", "75th Percentile"],
    "Salary": [
        us_senior_2022["salary_in_usd"].quantile(.25),
        us_senior_2022["salary_in_usd"].median(),
        us_senior_2022["salary_in_usd"].mean(),
        us_senior_2022["salary_in_usd"].quantile(.75) ]})

sns.barplot(
    x="Benchmark",
    y="Salary",
    data=salary_benchmarks)

plt.title("2022 Senior US Data Scientist Market")
plt.xlabel("Salary Benchmark")
plt.ylabel("Salary in USD")
plt.show()
```



Recommendations:

Our organization should utilize \$150-205k range to hire Sr. level candidate that could lead a team of data scientists. To be competitive, we could go above the median range and stay slightly under the upper, 75%, and negotiate from \$160,000 - \$190,000.